

# Lekkala Ganesh

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## PROFESSIONAL SUMMARY

Full-stack developer with experience across four professional roles spanning AI-powered product development, healthcare UI engineering, and scalable web platforms. Skilled in AI integration (LLM orchestration, multi-provider fallback systems), VRAM optimization, and building accessible (WCAG 2.1 AA), secure (RLS, JWT) products with React, Next.js, and FastAPI. Seeking software engineering roles in Bangalore.

## TECHNICAL SKILLS

**AI & ML:** PyTorch, Hugging Face (Diffusers, Transformers), 4-bit Quantization (bitsandbytes), CUDA, Scikit-learn, LLM API Integration (GPT-4o, Gemini), Model Fallback Architectures

**Backend & Data:** REST/WebSocket APIs, FastAPI, Node.js, PostgreSQL (Row-Level Security), Supabase, MongoDB Atlas

**Frontend:** React, Next.js (SSR/SSG), TypeScript, Tailwind CSS, Framer Motion, Three.js

**DevOps & Tools:** Docker, GitHub Actions (CI/CD), Linux Shell Scripting, Vercel, Git

**Testing & Accessibility:** Vitest, Playwright (E2E), WCAG 2.1 AA, SEO (JSON-LD)

## PROFESSIONAL EXPERIENCE

### Jr. Software Developer

AlterSquare

Remote

Dec 2025 – Mar 2026

- **Full-Stack Delivery:** Owned the end-to-end development of 9 client-facing web applications, delivering all projects from requirement gathering to production within 90 days with zero critical bugs reported post-deployment.
- **AI System Integration:** Engineered a multi-provider AI fallback system (GPT-4o, Gemini, OpenRouter) for high availability across apps, including real-time SSE voice streaming (VoxAI) and 3D shadow simulations (SunScope Pro).
- **Security & Scalability:** Architected high-security backends using Row-Level Security (RLS) and JWT authentication on Supabase/PostgreSQL, managing secure data for financial ([Finance Tracker](#)) and contractor platforms ([2A Construction](#)).
- **Accessibility:** Achieved WCAG 2.1 AA compliance across all products using custom a11y hooks and automated Playwright testing, expanding the accessible user base for client startups.

### Web Developer

Dash Technologies Inc.

Ahmedabad, Gujarat

Jan 2025 – Jul 2025

- **Healthcare UI Delivery:** Built 12+ modular, responsive React components for a patient-facing portal, reducing accessibility audit findings by 40% through semantic HTML and HIPAA-aligned data handling.
- **Cross-Browser Engineering:** Ensured 100% visual parity across Chrome, Firefox, Safari, and Edge for complex modules like appointment scheduling and patient history.

### Full Stack Web Development Intern

Slash Mark IT Solutions

Remote

Dec 2024 – Apr 2025

- Delivered 6 full-stack projects across 3 tiers (basic to advanced) including a real-time Weather Forecast app with API integration, a URL Shortener API (Node.js, Express, MongoDB), and a Food Delivery platform with authentication, geolocation, and payment flow using React and Node.js.

### Machine Learning Intern

Bharat Intern

Virtual

Oct 2023 – Nov 2023

- Built regression and classification models (House Price Prediction, Wine Quality Prediction, Iris Classification) using Scikit-learn and Pandas in Jupyter notebooks, applying feature engineering, cross-validation, and hyperparameter tuning.

## KEY TECHNICAL PROJECTS

**AuraGen – AI Image & Video Generation Platform** | PyTorch, HF Diffusers, FastAPI, Next.js, CUDA

[GitHub](#)

- **VRAM Optimization:** Engineered a specialized 4GB VRAM optimization stack (NF4 quantization, attention slicing, VAE tiling) that reduced model memory requirements from 12.8GB to 3.2GB, enabling text-to-video (Wan 2.1) on consumer GPUs.
- **System Orchestration:** Designed a real-time job queue with WebSocket progress streaming and hybrid cloud burst routing (HuggingFace Inference API fallback) to handle peak inference loads.

**GoVault – AI-Powered Travel Organizer** | Next.js, TypeScript, Supabase, GPT-4o, Playwright

[GitHub](#)

- **AI-Powered Import:** Built a GPT-4o integration that extracts structured booking data from pasted emails and receipts, with offline-first architecture using IndexedDB and background sync.
- **Full Travel Suite:** Implemented day-by-day itinerary builder, multi-currency expense tracking, family traveler profiles, and collaborative trip sharing with privacy controls.

**SymptomChecker AI – ML-Powered Health Assistant** | Flask, Scikit-learn, React, MongoDB, JWT

[GitHub](#)

- **Model Accuracy:** Built a Scikit-learn prediction engine achieving 95%+ accuracy across 40+ medical conditions, integrated with location-based hospital recommendation APIs.

## EDUCATION

Parul University

Bachelor of Technology in Computer Science & Engineering; GPA: 7.53/10

Vadodara, Gujarat

Jul 2022 – Jun 2025